

Chia-Yuan Wu

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EDUCATION

Lehigh University (LU) <i>Ph.D.</i> in Industrial and Systems Engineering National Chiao Tung University (NCTU) <i>M.S.</i> in Industrial Engineering and Management <i>B.B.A.</i> in Management Science	Bethlehem, PA <i>Aug. 2022 - Present</i> Hsinchu, Taiwan <i>Sept. 2015 - Jul. 2017</i> <i>Sept. 2011 - Jun. 2015</i>
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WORK EXPERIENCE

Advanced Micro Devices, Inc. (AMD) <i>Software System Designer</i> - Developed and optimized a unified data feed solution using MSSQL, C#, and VSTO to automate SCM and S&OP processes, reducing user workload by 50% and eliminating manual scheduling - Customized and enhanced Kinaxis RapidResponse functionalities via JIRA to improve supply chain planning modules and decision-making automation, reducing user support load and streamlining planning workflows - Delivered scalable integration of GPU/CPU planners by collaborating with cross-functional teams, managing concurrent tasks across supply chain modules, improving coordination efficiency and execution speed	Hsinchu, Taiwan <i>Apr. 2021 - Jul. 2022</i>
AU Optronics Corp. (AUO) <i>Senior Software Engineer</i> - Led a 3-member cross-functional team to develop MIP-based scheduling and resource optimization systems using IBM CPLEX, PL/SQL, and C#, reducing operational time by 90% and replacing legacy rule-based heuristics - Built and deployed an end-to-end optimization pipeline integrated with SQL Server and Oracle DB, boosting scheduling precision and reducing production delays across multiple lines - Engineered an inventory pegging optimization system with MIP models, increasing order success rate by 2× and cutting buyer workload by 95% - Designed interactive dashboards for daily/weekly production monitoring and KPI tracking using VSTO and ASP.NET, enabling real-time decision-making	Hsinchu, Taiwan <i>Sept. 2017 - Apr. 2021</i>

RESEARCH PROJECTS

Synthetic Data for Mitigating Unfairness in Collaborative Machine Learning <i>LU Research Assistant Advisors: Prof. Frank Curtis, and Prof. Daniel Robinson</i> - Implemented a machine learning framework combining synthetic data generation and differential privacy to mitigate algorithmic bias and privacy risks in collaborative learning systems using Python, PyTorch, and MySQL - Formulated a novel bilevel optimization model integrating data distillation and unfairness mitigation to generate representative synthetic datasets, preserving client privacy and enabling fair, robust server-side training - Architected a single-round distributed learning system that reduces client-server communication, enhancing both computational efficiency and prediction fairness across decentralized networks	Bethlehem, PA <i>Jan. 2024 - Jun. 2025</i>
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PUBLICATIONS (FULL LIST)

[1] **Chia-Yuan Wu**, Frank E. Curtis, and Daniel P. Robinson. (2025). "A Bilevel Optimization Approach for Computing Synthetic Data to Mitigate Unfairness in Collaborative Machine Learning." *Preprint in review.*

CONFERENCE PRESENTATIONS

- A Bilevel Optimization Method Using Synthetic Data to Mitigate Unfairness and Preserve Privacy in Collaborative Machine Learning. Oral Presentation (upcoming), *2025 INFORMS Annual Meeting*, Atlanta, GA, October 2025.
- Enhancing Fairness in Machine Learning through Training Synthetic Datasets in Multi-Client Scenarios. Oral Presentation, *MOPTA 2024*, Bethlehem, PA, August 2024.

TECHNICAL SKILLS

Programming Languages: Python, C#, PL/SQL, T-SQL, Matlab
Database Management Systems: MySQL, SQL Server, Oracle
AI Tools & Optimization Software: PyTorch, TensorFlow, Keras, scikit-learn, CPLEX, GLPK, AMPL, Gurobi
Other Libraries & Toolkits: Pandas, Numpy, ASP.NET, VSTO, Git, CI/CD, Microsoft Office, L^AT_EX